

McHenry County

RACES / ARES / STARCOM

FIELDCOMMS

EmComm Field Server v1.0

USER GUIDE

■ Net Control

■ ICS Platform

■ APRS Tactical
Map

■ Kiwix Library

■ Weather Alerts

■ Repeater
Database

June 21, 2026 · For Authorized Operator Use Only

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1

GETTING STARTED

Connecting to FieldComms

1. Getting Started — Connecting to FieldComms

FieldComms is a self-contained emergency communications server running on a Raspberry Pi. It broadcasts its own WiFi network called **EMCOMM-NET**. Any smartphone, tablet, or laptop can connect and access all tools through a standard web browser — no app installation required, no internet needed.

Step 1 — Connect to EMMCOMM-NET WiFi

FIELD	VALUE
WiFi Network Name (SSID)	EMCOMM-NET
Password	(provided by your net manager)
Security	WPA2
Your device will receive IP	192.168.50.100 – 192.168.50.200

Step 2 — Open the Dashboard



Open any web browser on your device and navigate to: <http://192.168.50.1> — The FieldComms dashboard will load. You can also try <http://fieldcomms.local> or <http://emcomm.local> as friendly aliases on most devices.

What You Can Access

URL / ADDRESS	WHAT IT OPENS
http://192.168.50.1/	Main FieldComms dashboard and all pages
http://192.168.50.1:8081/	Kiwix offline library (WikiMed, Wikipedia, iFixit...)
http://192.168.50.1:8080/	Graywolf APRS client interface
http://192.168.50.1:8090/	Pat Winlink email-over-radio
http://fieldcomms.local/	Alternate friendly address (same as above)

NOTE

Internet is NOT required. All core FieldComms features work offline. When Ethernet internet is connected to the Pi, bonus features activate automatically: live NWS weather alerts, HF propagation solar data, and APRS-IS internet gateway access.

2

MAIN DASHBOARD

http://192.168.50.1/

2. Main Dashboard

The dashboard is your central hub. It shows live system status, active NWS weather alerts, a live APRS station table, and links to every tool.

Three Dashboard Modes

Select a mode from the mode bar at the top. The dashboard remembers your last-used mode.

MODE	BUTTON	TOOLS AVAILABLE
Amateur Radio	■	Net Control Logger, Observer Mode, Callsign Lookup, JS8Call, APRS Tactical Map, HF Propagation, Repeaters, Grid Square, NTS Radiogram, ICS-213, ICS-214, Dead Man's Switch, Roster, Pre-Flight, Cheat Sheets, Print Center, Reference Library
Starcom / Public Safety	■	Starcom Net Logger, Weather Net, SAR Net, Observer Mode, Resource Tracking Map, Resource Board, Member Roster, Facilities Directory, ICS Platform, ICS-213, ICS-214, Print Center
ICS Mode	■	ICS Platform (Command, Operations, Planning, Logistics, Finance, Planning P), Tactical Map, Resource Map, Resource Board, Roster, Facilities, ICS-213, ICS-214, ICS-309, Winlink Form Import, Repeaters, Cheat Sheets, Reference Library

Dashboard Elements

ELEMENT	DESCRIPTION
Hero Bar	Station callsign badge, system name, live UTC and local 24-hour clocks updated every second
Mode Bar	Three mode buttons (Amateur Radio / Starcom / ICS) — click to switch dashboard view
■ NWS Weather Alerts	Live National Weather Service alerts color-coded by severity. Click any alert to expand full details, onset time, expiry countdown, and protective action instructions.
Operation Cards	Quick-launch tiles for all tools in the current mode. Each card shows the tool name, description, and the port or API it connects to.
■ APRS Station Table	Live table of APRS stations heard by Graywolf. Shows callsign, distance, course/speed, comment, and last-heard time. Updates every 30 seconds.
Status Sidebar	System health (CPU, memory, disk, temperature), service status dots (green=running, red=stopped), GPS status, and Dead Man's Switch state.

3

AMATEUR NET CONTROL LOGGER

<http://192.168.50.1/netcontrol.html>

3. Amateur Net Control Logger

The Net Control logger is the primary tool for running ARES/RACES amateur radio nets. It supports multiple simultaneous nets, automatically looks up FCC callsign data, and exports logs in ICS-309 format.

Starting a Net

When you first open Net Control, create a new net session:

1. Click `[+ New Net]`
2. Enter Net Name → e.g. "McHenry County ARES Thursday Evening Net"
3. Enter Frequency → e.g. "147.015 MHz / 107.2 Hz"
4. Enter Net Control → your callsign
5. Click `[Activate Net]`

→ The net is now live. Share the observer link for read-only viewers.

Logging Check-Ins

Each row in the log represents one check-in. The FCC database auto-fills operator name and location when you type a callsign.

Net Control Logger — Check-In Entry

Callsign	W9XYZ (type → auto-fill fires)
Name	Smith, John (auto-filled from FCC DB)
Location / Grid	CRYSTAL LAKE IL (auto-filled)
Precedence	ROUTINE / WELFARE / PRIORITY / EMERGENCY
Traffic	NTS message count or remarks
Timestamp	14:22:05 UTC (auto-set on log click)

TIP Walk-in stations not in the FCC database get a WALK-IN badge. You can manually enter their name and promote them to the roster later.

Precedence Levels

PRECEDENCE	COLOR	MEANING
EMERGENCY	Red	Immediate threat to life. Takes absolute priority over all traffic.
PRIORITY	Amber	Serious urgency — critical supply requests, official needs.
WELFARE	Blue	Health and welfare inquiries from or to disaster areas.
ROUTINE	Normal	Standard traffic. Processed in order during normal net cycles.

Net Control Features

- **FCC Auto-Fill** — Type any US amateur callsign — name and location fill automatically from the local FCC database. No internet required.
- **Multiple Nets** — Run several nets simultaneously. Each has its own log, frequency, and NCS. Switch between them with the net selector tabs.
- **Observer Link** — Click the share icon to copy a URL like /observer.html?net=Thursday-Net. Anyone can view the live log read-only.
- **Server Sync** — Logs are saved to the FieldComms server. They persist through page refreshes and can be loaded on any device on EMCOMM-NET.
- **ICS-309 Export** — Export the complete net log as a formatted ICS-309 Communications Log — ready to print or include in an IAP.
- **JSON Backup** — Download the raw log data as JSON for archiving or importing into another system.
- **DMS Integration** — The Dead Man's Switch watches this net. Inactivity beyond the threshold triggers a warning then alarm state.

4

STARCOM NET LOGGER

<http://192.168.50.1/starcom.html>

4. Starcom Net Logger

The Starcom logger is purpose-built for public safety Starcom radio net logging. It identifies units by Radio ID and unit number rather than amateur callsigns, and uses the **sc-** prefix convention for all net names.

Key Differences from Amateur Net Logger

FEATURE	AMATEUR NET CONTROL	STARCOM LOGGER
Unit identification	FCC Callsign (e.g. W9XYZ)	Radio ID / Unit number (e.g. 1234)
Net name prefix	Free form	sc- prefix (e.g. sc-McHenry-TAC1)
ID lookup	FCC database auto-fill	Unit name / callsign manual entry
Dispatch field	Not applicable	Dispatch center name field included
Export format	ICS-309	ICS-309 (same format, Starcom header)

Running Multiple Simultaneous Nets

Like the Amateur Net Control Logger, the Starcom logger fully supports multiple simultaneous nets. This is essential during large activations where you run a Starcom General Net, a Weather Net, and a SAR Net at the same time.

- Click **+ New Net** for each net you want to open.
- Each open net appears as a clickable badge in the active nets panel on the left.
- Click any badge to switch to that net — each has its own independent check-in and traffic log.
- Typical multi-net activation: Starcom General + Weather Net + SAR Net all open simultaneously.

TIP Net Control switches between active nets by clicking the net badges in the left panel. Each net's check-in count is shown on its badge so you can see at a glance which nets are active and how many units are checked in to each.

Starcom Check-In Fields

Starcom Net Logger — Entry

Radio ID / Unit #	1234 (Starcom unit identifier)
Unit Name / Callsign	MCHEMRY CO ARES or W9XYZ
Channel / Frequency	TAC-1 / 155.340 MHz
Location	WOODSTOCK EOC
Remarks	Requesting traffic or other notes
Time	15:44:30 UTC (auto-stamped)

NOTE Starcom nets are distinguished from amateur nets in all exports and on the Dead Man's Switch by the sc- prefix. Keep this prefix on all Starcom net names to ensure proper separation in reports and the DMS monitor.

5

NET OBSERVER

<http://192.168.50.1/observer.html?net=NETNAME>

5. Net Observer (Read-Only View)

The observer page provides a read-only, auto-refreshing view of any active net. Share the link with served agency personnel, EOC staff, or section leaders who need to monitor traffic without touching the net control interface.

TIP

The net name is passed in the URL: <http://192.168.50.1/observer.html?net=Thursday-Net> — The page refreshes automatically every 15 seconds. Bookmark it for quick access. Anyone on EMCOMM-NET can open this link — no login required.

6

MEMBER ROSTER<http://192.168.50.1/roster.html>

6. Member Roster

The Member Roster is the authoritative directory for McHenry County RACES/ARES/Starcom personnel. It tracks certifications, equipment capabilities, contact information, and deployment activations.

Roster Tabs

TAB	CONTENTS
Directory	Full member list with callsign, name, role, phone, email, grid square, certification dots, and equipment badges. Filter by role or certification.
Activations	Track which members are currently deployed, their assigned role, check-in time, and location. Promote walk-in check-ins from net control.

Tracked Certifications (11 total)

ICS-100	ICS-200	ICS-300	ICS-400	ICS-700	ICS-800
EmComm I	EmComm II	CPR/AED	First Aid	CERT	

Tracked Equipment (13 capabilities)

HF	VHF	Digital	Packet	PACTOR	VARA HF	VARA FM
APRS	Winlink	Go-Box	Generator	Battery	Vehicle	

Member Roles

ROLE	DESCRIPTION
Net Control (NCS)	Primary net controller — manages check-ins and traffic flow
Operator	Standard ARES/RACES field operator
Liaison	Agency liaison — interface between amateur operators and served agency
Emergency Coordinator (EC)	Oversees ARES/RACES operations for the county

Importing & Exporting

- **CSV Import** — import a spreadsheet of members. Columns: callsign, first_name, last_name, role, phone, email, grid.
- **CSV Export** — download the full roster as a spreadsheet for backup or sharing with ARRL/ARES leadership.

7

RESOURCE BOARD<http://192.168.50.1/resources.html>

7. Resource Board

Track all personnel, vehicles, and equipment assigned to an activation. Each resource card shows current status and can be cycled through states with a single click.

Resource Status States

STATUS	COLOR	MEANING
Available	Green	Resource is on scene and available for assignment
Assigned	Amber	Resource is actively tasked with a specific assignment
Staging	Blue	Resource is en route or at the staging area, not yet deployed
Out of Service	Red	Resource is unavailable — mechanical, medical, or administrative reason
Demobilized	Grey	Resource has been released from the incident

Resource Types

- Personnel (individuals, teams)
- Vehicles (go-box trucks, command vehicles)
- Equipment (generators, repeaters, antennas, go-boxes)
- Facilities (EOC, staging areas, shelters)

TIP

Click any resource card's status badge to cycle it to the next state. Use the CSV Import to bulk-load resources at the start of an activation. Stats strip at the top shows total counts by status at a glance.

8

TACTICAL APRS MAP<http://192.168.50.1/tactical.html>

8. Tactical APRS Map

The tactical map displays live APRS station positions on an interactive Leaflet map. It pulls data from two simultaneous sources — Graywolf TNC and YAAC — merging them into a single unified picture. APRS symbols, callsign labels, and station trails are all shown.

Sidebar Tabs

TAB	CONTENTS
Stations	List of all heard APRS stations, sorted by most recent. Filter by source (Graywolf / YAAC / APRS-IS), band type, or search by callsign. Click any station to pan the map to it.
Messages	Incoming and outgoing APRS messages. Send a message to any callsign via Graywolf or YAAC directly from this panel.
Markers	Place manual overlay markers (EOC, shelters, command posts, staging areas, repeaters, hazmat zones). Click "Pick on Map" then click any map location.
Sources	Live status of Graywolf (port 8080), YAAC (port 8082), and APRS-IS. Shows station count per source, WebSocket connection state, and merge statistics (stations seen by both vs. each alone).
Settings ■	Station callsign, coordinates, refresh rate, map tile source, callsign label on/off, station age threshold, range ring radius.

Map Layer Controls

BUTTON	WHAT IT SHOWS
■ Graywolf	Stations heard by the Graywolf TNC client (green border on icon)
■ YAAC	Stations heard by YAAC (blue border on icon)
■ APRS-IS	Internet gateway stations (purple border) — requires internet + aprs.fi API key
■ Overlays	Manual markers — EOC, shelters, command posts, etc.
■ My Station	Your station's position (blue dot with callsign label)
■ Range Ring	Dashed circle showing your configured coverage radius
■ Track	Track a selected station's movement history — draws path as it moves

Station Icon Colors

FILL COLOR	MEANING
Green (bright)	Heard within the last 15 minutes (fresh)
Amber	Heard 15–60 minutes ago
Grey	Heard more than 60 minutes ago (stale)
Blue (dot)	Your own station marker

Purple (dot)

Manual overlay marker

TIP

Click any station icon on the map to see a popup with full detail: APRS symbol, type, comment, speed, course, altitude, path, and last-heard time. The popup also has Track and Message buttons for quick actions.

9

STARCOM RESOURCE MAP

<http://192.168.50.1/resmap.html>

9. Starcom Resource Map

The Starcom Resource Map is a Leaflet-based map for manually placing and tracking Starcom units. Unlike the APRS tactical map, positions here are placed manually by the operator. It supports unit placement, status color coding, zone drawing, and polygon area markup for tactical overlays.

TIP

Use the Starcom Resource Map alongside the Starcom Net Logger during a public safety activation. Place units on the map as they check in, and update their status as assignments change.

10

FCC CALLSIGN LOOKUP<http://192.168.50.1/callsign.html>

10. FCC Callsign Lookup

Search the local FCC amateur license database — the entire US amateur callsign database (~800,000 licensees) stored offline on the Pi. No internet required. Results include licensee name, address, license class, grant date, expiry date, and trustee callsign for club licenses.

TIP

Direct URL lookup: <http://192.168.50.1/callsign.html?call=W9XYZ> — Useful for bookmarking specific calls or linking from net control exports.

11

DEAD MAN'S SWITCH MONITOR

<http://192.168.50.1/deadmans.html>

11. Dead Man's Switch Monitor

The Dead Man's Switch (DMS) monitors each active net for inactivity. If a net goes silent beyond a configured threshold, it escalates through warning and trigger states — alerting operators that the net may have lost communications.

STATE	COLOR	MEANING	ACTION
DISARMED	Grey	Net not being monitored	Click ARM to activate monitoring
ARMED	Green	Monitoring active, net is communicating	Normal operations. Reset timer on each check-in.
WARNING	Amber (pulsing)	75% of inactivity threshold reached	Contact net control. Resume net activity to clear.
TRIGGERED	Red (flashing)	Inactivity threshold exceeded	Immediate action required. Investigate net status.

NOTE The DMS page shows a countdown timer for each armed net. The ARM button activates monitoring; RESET clears the timer when the net resumes. DISARM stops monitoring entirely. The dashboard also shows DMS state in the sidebar.

12

PRE-FLIGHT DEPLOYMENT CHECKLIST

<http://192.168.50.1/preflight.html>

12. Pre-Flight Deployment Checklist

The pre-flight checklist is a structured GO / CAUTION / NO-GO readiness assessment to complete before activating for any deployment. It covers all critical systems and is saved as a timestamped JSON record.

Checklist Sections (~35 items total)

SECTION	ITEMS CHECKED
■ Power Systems	Shore/generator power, battery backup, UPS, fuel supply, runtime estimate
■ Communications Equipment	HF radio, VHF/UHF radio, TNC, antennas, coax, SWR check
■ Computing & Software	Pi boot, all FieldComms services green, Kiwix, FCC DB, browser access
■ Personnel & Staffing	NCS identified, EC notified, backup NCS available, RACES/ARES activation
■ Logistics & Supplies	Food and water (72 hr), consumables, first aid kit, backup equipment
■ Safety & Security	Lightning protection, antenna grounds, safe working conditions
■ Agency Coordination	EOC check-in, served agency liaison contacted, ICS position filled

GO / CAUTION / NO-GO System

- **Required items** must be checked for a GO result. Any unchecked required item produces CAUTION or NO-GO.
- **Optional items** are best practice but do not block a GO result.
- Export your completed checklist as a JSON file for the incident record.

13

ICS PLATFORM — OVERVIEW

<http://192.168.50.1/ics/>

13. ICS Platform — Overview

The ICS Platform provides a complete Incident Command System management interface organized into the five standard ICS sections. Each section has its own page with appropriate forms and tracking tools. All data is saved to the FieldComms server and accessible from any device on EMCOMM-NET.

Creating an Incident

1. Open <http://192.168.50.1/ics/>
2. Click [+ New Incident]
3. Fill in:
 - Incident Name → e.g. "Severe Storm 2026-06"
 - Type → Natural Disaster / HazMat / SAR / Mass Casualty / etc.
 - Jurisdiction → e.g. McHenry County
 - Incident Commander
 - Op Period Duration → 12 hours (default)
 - Location
 - Situation Summary
4. Click [Activate Incident]

→ All five ICS sections are now active with the incident loaded.

ICS Navigation Bar

SECTION	COLOR	KEY FUNCTIONS
■ Overview	Gold	Incident selector, section stats, activity feed, advance period button
■ Command	Gold	Incident objectives (ICS-202), command staff (ICS-203), situation summary (ICS-201), safety hazards
■ Operations	Red	T-card board, resource list, assignment tracking (ICS-204)
■ Planning	Blue	Situation status (ICS-209), IAP form tracker, resource status table, documentation log
■ Logistics	Green	Comms plan (ICS-205), supply tracking, facilities, meal tracking, check-in list (ICS-211)
■ Finance/Admin	Purple	Cost accounting, time tracking, procurement orders, compensation and claims

NOTE The [■ Advance Period] button on the ICS Overview closes the current operational period and opens the next one. All forms and assignments carry forward. Use this at each operational period transition.

14

ICS COMMAND<http://192.168.50.1/ics/command.html>

14. ICS Command Section

FEATURE	HOW TO USE IT
Incident Header	Incident name, number, operational period, IC, location, prepared-by. Pre-populated from the incident you activated.
Incident Objectives (ICS-202)	Enter numbered tactical objectives for the current period. Add with the text box and + Add button. Delete individual objectives with X.
Safety Message	Free-text safety message for the period — weather, hazards, PPE requirements.
Weather Summary	Current conditions and forecast for the operational area.
Command Staff (ICS-203)	Fill in name and agency for each ICS position: IC, Safety Officer, PIO, Liaison Officer, and all four section chiefs.
Situation Summary (ICS-201)	Current situation narrative, accomplishments this period, and planned actions for the next period.
Safety Hazards Log	Running log of identified hazards with ■ badge. Add hazards with the text box at the bottom.

TIP Print ICS-201 using the ■ Print ICS-201 button. The page renders a formatted print view.

15

ICS OPERATIONS<http://192.168.50.1/ics/operations.html>

15. ICS Operations Section

FEATURE	HOW TO USE IT
T-Card Board	Visual Kanban-style board with four columns: Available / Staged / Assigned / Out of Service. Drag-click cards to move resources between columns. Click any card to view detail and change assignment.
Resource List	Table view of all resources with ID, type, name, capability, status, assignment, and contact. Click any row to edit.
Assignments (ICS-204)	List of all resources currently in Assigned status with their tasking and contact info.

TIP Add resources with the [+ Add Resource] button. Set Resource ID (e.g. E-101), type, capability, and initial status.

16

ICS PLANNING

<http://192.168.50.1/ics/planning.html>

16. ICS Planning Section

FEATURE	HOW TO USE IT
Situation Status (ICS-209)	Status summary cards: total personnel, resources, assigned, available, out of service, and current period. Fill in incident status, % contained, situation narrative, significant events, weather, and agency list.
IAP Forms	Checklist of all ICS forms with status indicators and links to open each one. Tracks which forms are complete for the current IAP package.
Resource Status Table	Grid showing how many of each resource kind are ordered, on scene, assigned, available, out of service, and still needed.
Documentation	Running log of all documents produced during the incident — forms, maps, photos, reports. Add entries with type, author, and notes.

TIP

The Planning P timing section lets you record key planning meeting times for the period.

17

ICS LOGISTICS<http://192.168.50.1/ics/logistics.html>

17. ICS Logistics Section

FEATURE	HOW TO USE IT
Comms Plan (ICS-205)	Build the radio communications plan. Each row is a channel: function/assignment, channel name, frequency, CTCSS tone, mode, and remarks. Default channels are pre-filled for McHenry County.
Supply Tracking	Log supply requests with item, category, quantity needed, quantity on hand, and priority. Visual progress bar shows fill rate.
Facilities	ICP location, base/camp details, staging area manager, and medical aid station information.
Food / Medical (ICS-206)	Meal service log (meal type, count served, menu) and medical unit leader, ambulance service, nearest hospital.
Check-In List (ICS-211)	Personnel check-in log with name/callsign, agency, ICS position, and check-in timestamp.

TIP

The Communications Unit Leader and ordering contacts are at the bottom of the Comms Plan tab.

18

ICS ADMIN<http://192.168.50.1/ics/finance.html>

18. ICS Finance / Admin Section

FEATURE	HOW TO USE IT
Cost Accounting	Log all expenditures: category (personnel, equipment, supply, food, transport), amount, description, vendor, and who approved it. Running total shown.
Time Tracking	Log person-hours for each operator: name, position, agency, hours worked, and hourly rate (0 = volunteer). Used for reimbursement claims.
Procurement	Track purchase orders: item, vendor, PO number, amount, and status (Requested / Ordered / Delivered / Closed).
Admin	Finance/Admin Section Chief info, billing agency, cost share type, claim deadline, and supporting documentation requirements.

TIPExport all finance data to CSV with the  CSV button for agency reporting.

19

NTS RADIOGRAM GENERATOR<http://192.168.50.1/nts.html>

19. NTS Radiogram Generator

Generate properly formatted ARRL National Traffic System radiograms. The form produces the standard preamble, address block, and text section. Completed radiograms are saved in a traffic log.

Radiogram Fields

FIELD	DESCRIPTION	EXAMPLE
Message Number	Sequential message number	001
Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE	ROUTINE
Handling Instructions	HX codes — delivery special instructions	HXC
Station of Origin	Originating station callsign	W9XYZ
Check (word count)	Word count of text — auto-calculated	22
Place of Origin	City and state of originating station	WOODSTOCK IL
Time Filed	UTC time — auto-filled	1422Z
Date Filed	UTC date	7 JUN
Addressee	Full name of recipient	JOHN SMITH
Address	Street address, city, state, ZIP	123 MAIN ST CHICAGO IL
Phone	Recipient phone number	312-555-0100
Text	Message body — ARRL standard all-caps	ARRIVED SAFELY...
Signature	Sender name	JANE SMITH

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ICS-213 GENERAL MESSAGE

<http://192.168.50.1/ics213.html>

20. ICS-213 General Message

The ICS-213 is the standard written message form for inter-agency and intra-agency communications on an incident. Fill in the header (incident name, to/from, position/agency, date/time, subject) and the message body. The page renders a print-formatted version — use the ■ Print button to produce a field-ready paper copy.

NOTE ICS-213 is for formal written communications. For routine spoken radio traffic, use the net control log. Use 213 when you need a written record of requests, situation updates, or resource orders.

21. ICS-214 Activity Log

The ICS-214 is a unit-level activity log required for every ICS section and unit. Enter the incident name, operational period, unit name, and unit leader. Add personnel to the table. Then log timestamped activity entries throughout the period. Print a formatted ICS-214 with all entries for the incident record.

22

HF PROPAGATION<http://192.168.50.1/propagation.html>

22. HF Propagation

The propagation page fetches live solar and geomagnetic data from hamqsl.com every 15 minutes when internet is available. It uses this data to estimate current HF band conditions.

INDICATOR	WHAT IT MEANS FOR HF
SFI (Solar Flux Index)	Higher = better high-band propagation. SFI >130: excellent. SFI 90-130: good. SFI <90: 40m/80m only.
Sunspot Number (SN)	More sunspots = more solar activity = better HF. Follows 11-year cycle.
A-Index	Daily geomagnetic activity. A <10: quiet (good HF). A 10-30: unsettled. A >30: disturbed.
K-Index (0-9)	3-hour geomagnetic snapshot. K ≤2: quiet. K 3-4: unsettled. K ≥5: storm — HF degraded. K ≥7: severe.
X-Ray Flux Class	Solar flare class. A/B/C: minor. M: moderate SID possible. X: strong HF blackout risk.
Band Condition Bars	Green=Good / Amber=Fair / Red=Poor for each HF band, both day and night conditions.
MUF / LUF Estimates	Maximum and minimum usable frequencies for local, regional, DX, and long-path circuits from EN90.

NOTE For EmComm: 40m is your backbone band — reliable day and night for regional to national paths. 80m excels at night for local/regional. The five 60m channels (5330.5–5403.5 kHz USB) are the interoperability standard for served agency communication. Monitor 14.300 MHz for IARU distress.

23

REPEATER DATABASE<http://192.168.50.1/repeaters.html>

23. Repeater Database

The Repeater Database displays RepeaterBook data with rich filtering. It works in three modes: offline file (primary), server API, or demo data.

SOURCE TAB	HOW TO USE
■ Offline File	Load a RepeaterBook CSV or JSON export. Drag-and-drop the file or click Browse. Data persists in the browser — no re-import needed after the first load. To get the file: repeaterbook.com → search your area → Export → CSV.
■ Server API	Pulls from the FieldComms server if <code>fetch_repeaters.py</code> has been run to download RepeaterBook data for your state.
■ Demo Data	Ten pre-loaded Central Ohio repeaters including ARES/SKYWARN examples.

Filter Options

- Band (10m / 6m / 2m / 1.25m / 70cm / 33cm / 23cm)
- Mode (FM Analog / D-STAR / C4FM-Fusion / DMR / P25)
- Status (On-Air / Off-Air)
- State (populated from loaded data)
- Use (Open / Closed / Private)
- EmComm (ARES / SKYWARN / RACES)
- Distance sort — nearest first or farthest first from your station coordinates

24

FACILITIES DIRECTORY

<http://192.168.50.1/facilities.html>

24. Facilities Directory

Maintain a directory of EOC locations, hospitals, shelters, staging areas, and command posts. Each entry stores address, coordinates, radio frequencies, CTCSS tone, contact person, callsign on site, generator status, ADA access, capacity, and operational notes.

TIP

Four default facilities are pre-loaded for McHenry County configuration. Edit these with your actual EOC, shelter, and staging area information. Click any card to expand full detail. Use the  Copy Address button to copy the address to the clipboard.

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GRID SQUARE CALCULATOR

<http://192.168.50.1/grid.html>

25. Grid Square Calculator

Convert between decimal latitude/longitude and Maidenhead grid squares in both directions. Supports 4-character (e.g. EN52) and 6-character (e.g. EN52ab) precision. Also calculates distance and bearing between any two grid squares. The canvas map shows the grid square highlighted on a North America outline.

RADIO CHEAT SHEETS<http://192.168.50.1/cheatsheets.html>

26. Radio Cheat Sheets

TAB	CONTENTS
Phonetic Alphabet	NATO phonetic alphabet A–Z with pronunciation guide, plus ITU number pronunciation (Zero, WUN, TOO, TREE, FOW-er, FIFE, SIX, SEV-en, AIT, NIN-er)
Q-Codes	Common amateur Q-codes (QRM, QRN, QRP, QSL, QTH, etc.) and EmComm net Q-codes (QNI, QNS, QND, QNN, etc.)
Prowords	Full procedure word reference — ROGER, WILCO, SAY AGAIN, BREAK, OVER, OUT, CORRECTION, SILENCE, AUTHENTICATE, and 25 others with meanings and example uses
Band Plan	2-meter and 70-cm band segments with frequencies and uses, HF emergency frequencies (ARES, ARRL, IARU, 60m channels), and special service bands (MURS, FRS, GMRS, Marine, Aviation)
NTS Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE definitions and ICS triage priority colors (Immediate / Delayed / Minor / Expectant)
ICS Structure	ICS org chart diagram and key ICS forms reference table (ICS-201 through ICS-309)
CTCSS / DCS	Full CTCSS tone table (67.0–250.3 Hz, 41 tones) and common DCS codes
Signal Reports	RST readability/signal/tone scale, APCO P25 BER quality table, FM voice report shorthand, and dBm to watts conversion table

27

PRINT CENTER<http://192.168.50.1/printcenter.html>

27. Print Center

The Print Center is a one-stop hub for all printable documents. It links to every printable form and includes a built-in incident cover sheet generator.

CATEGORY	AVAILABLE DOCUMENTS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, NTS Radiogram, Pre-Flight Checklist
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, CTCSS/DCS & Signal Reports
Operations	Net Control Log (ICS-309) — exported from net control, Starcom Net Log, Member Roster, Resource Board
Cover Sheet Generator	Fill in incident name, number, IC, operational period, frequency, location, and situation summary. Generates a formatted cover sheet for your IAP packet. Preview in-page, then print to a new window.

37

PRINTER SETUP

<http://192.168.50.1:631>

37. Printer Setup — Connecting a Printer to EMCOMM-NET

FieldComms has print buttons on 17 pages — all using the browser standard print function which sends the job to whatever printer the operator browser can reach. Three connection options are available:

OPTION	HOW IT WORKS	BEST FOR
A — Own printer	Each operator prints to their own locally connected printer. No Pi setup needed — works immediately.	Single operator or when every device already has a printer
B — USB printer via Pi (CUPS)	USB printer plugged into the Pi. CUPS shares it across EMCOMM-NET so every device can print to it. Admin at http://192.168.50.1:631 . Installed automatically by the FieldComms installer.	Field activations — one shared printer for all operators. Supports Windows, Mac, iOS (AirPrint), Android, Chromebook.
C — Network printer	Wi-Fi or Ethernet printer connected directly to EMCOMM-NET (via ASUS router or UniFi switch). No Pi configuration needed. Devices discover it automatically via Bonjour/mDNS.	When a Wi-Fi capable printer is available — simplest shared setup

Adding the CUPS Shared Printer on Your Device

- **Windows:** Settings → Printers & scanners → Add a printer. The CUPS printer appears automatically on EMCOMM-NET.
- **Mac:** System Settings → Printers & Scanners → + → select from Default tab (Bonjour discovery).
- **iPhone/iPad:** AirPrint — tap Share → Print in any FieldComms page, select the printer. No setup needed.
- **Android:** Install CUPS Print app → add printer at 192.168.50.1:631.
- **CUPS admin (add/manage printers):** <http://192.168.50.1:631> from any browser on EMCOMM-NET.

TIP

For most activations, Option B (USB printer via CUPS) is recommended. A Brother HL-L2350DW or HP LaserJet plugged into the Pi USB hub gives all operators shared access with no per-device setup. For mobile deployments, the Canon PIXMA TR150 or HP OfficeJet 200 are battery-powered options.

28

KIWIX OFFLINE LIBRARY<http://192.168.50.1:8081/>

28. Kiwix Offline Library

Kiwix provides a full offline web library served at port 8081. Every device on EMCOMM-NET can browse it — no internet required. Content is downloaded at install time or on demand by the net manager.

CONTENT	WHAT'S IN IT	BEST FOR
WikiMed Medical Encyclopedia	Symptoms, diagnoses, treatments, medications, procedures, anatomy	Mass casualty events, medical support operations
Wikipedia (Mini)	Compact English Wikipedia — facts, geography, science, history	General field reference, briefing preparation
Wikivoyage	Travel guides, emergency shelters, local resources, evacuation routes	Shelter-in-place planning, unfamiliar jurisdictions
Wikibooks	First aid, survival, ham radio, electronics, field skills how-to guides	Training reference, field skills refresher
iFixit Repair Manuals	Step-by-step repair guides for radios, computers, generators, medical equipment	Field equipment repair without internet access
Wikipedia Medicine (Full)	Complete clinical reference — diseases, drugs, surgery, pharmacology	Advanced medical support, extended operations
Wikiversity	ICS courses, CERT training, ham radio education, first responder materials	Training sessions, exercises, certification prep
Electronics Stack Exchange	Q&A; for radio circuits, antenna design, power systems	Technical troubleshooting, antenna problems

TIP

Kiwix has a full-text search engine built in. Type any medical symptom, procedure, or topic into the search bar and it searches across all installed content simultaneously. Results appear instantly — completely offline.

36. Network — EMCOMM-NET & AiMesh Coverage Extension

EMCOMM-NET is the Wi-Fi network broadcast by the ASUS RT-BE58 Go travel router. All FieldComms tools are accessible at **http://192.168.50.1** from any device connected to EMCOMM-NET. The Pi does not run its own Wi-Fi — the ASUS router handles all wireless access and DHCP for the 192.168.50.x subnet.

Single Router Coverage

A single ASUS RT-BE58 Go covers approximately 2,000–2,500 sq ft under typical indoor conditions. For most single-room EOC activations this is sufficient.

Extending Coverage — ASUS AiMesh

For larger deployments — multi-room EOC buildings, large shelters, outdoor staging areas, or multi-floor facilities — add one or more AiMesh nodes. AiMesh nodes are additional ASUS routers paired to the primary RT-BE58 Go. All nodes share the same EMCOMM-NET SSID and 192.168.50.x subnet. Devices roam automatically between nodes with no reconfiguration — the Pi remains at 192.168.50.1 regardless of which node a device connects through.

SCENARIO	SETUP
Single room EOC ($\leq 2,500$ sq ft)	Primary RT-BE58 Go only — no extension needed
Multi-room / large shelter (2,500–7,500 sq ft)	Primary + 1 AiMesh node (wireless or wired backhaul)
Large building or campus ($> 7,500$ sq ft)	Primary + 2–3 AiMesh nodes, wired Ethernet backhaul recommended
Outdoor SAR staging area	Primary at command post + battery-powered RT-BE58 Go nodes at field positions

- Any AiMesh-compatible ASUS router can be used as a node — the RT-BE58 Go (same as primary) is simplest.
- Pairing a new node takes under 5 minutes: reset the node, go to AiMesh → Add Node on the primary router, select the node, done.
- Wired Ethernet backhaul (CAT 6 from UniFi switch to node LAN port) is recommended for permanent setups — frees up Wi-Fi spectrum for clients.

TIP

See Chapter 34 of the User Manual (Section 33.8) for the full step-by-step AiMesh node setup procedure and compatible hardware list.

29

HEALTH MONITOR<http://192.168.50.1:5051/health>

29. Health Monitor

The health monitor runs as a background service (port 5051) and provides real-time system diagnostics. The dashboard sidebar displays key metrics. The raw JSON endpoint can be queried by any monitoring tool.

METRIC	NORMAL RANGE	ACTION IF EXCEEDED
CPU Temperature	< 70°C	Improve ventilation. Pi 5 throttles at 85°C.
Memory Usage	< 80%	Close unused browser tabs. Restart non-essential services.
Disk Usage	< 90%	Archive old logs. Remove unused ZIM files.
Service Status	All green dots	Red dot = service down. Restart: <code>sudo systemctl restart SERVICE</code>
Internet	Connected (when available)	Check Ethernet cable. All core features work offline.
GPS	Fix acquired (if GPS attached)	No GPS is fine — coordinates configured at install.

21

ICS-309 COMMUNICATIONS LOG

<http://192.168.50.1/ics309.html>

21. ICS-309 Communications Log

The ICS-309 captures a formal log of all message traffic during an activation. It is required for after-action reporting and is submitted with the incident record.

FIELD	WHAT TO ENTER
Incident Name	Name of the active incident (auto-populated if an incident is active)
Operational Period	Current period number and date range
Radio Operator	Your name and callsign
Station/Net	Net name or radio station identifier
Log Rows	Date/Time, From, To, Subject/Message summary — one row per message

How to Use

- Click **Add (timestamp now)** to add a row with the current UTC time stamped in.
- Fill in From, To, and the message Subject for each entry.
- Click **Generate Form** to render the printable ICS-309.
- Click **Save to Incident** to file it in the incident record.
- The Net Control and Starcom loggers can also export their traffic logs directly as ICS-309 format.

TIP

Link the ICS-309 to the active incident using the Incident dropdown at the top of the page. This files it permanently with the incident for after-action review.

31. Reference Library

The Reference Library stores all field reference documents — radio manuals, interoperability plans, ICS form packages, SOGs, agency plans, and training materials. Files are stored on the Pi and accessible from any device on EMCOMM-NET. The library accepts uploads up to 200 MB.

Uploading a Document

- Click **■ Upload Document**. A panel slides in from the right.
- Drag-and-drop the file or click Browse. Accepts PDF, Word, Excel, images, KML, ZIP, GPX.
- Fill in Title, Category, Source, Description, Tags, Revision, and Expiry (if relevant).
- Under **Show on tab(s)**, check Amateur Radio, ICS/EmMgmt, or both for cross-referencing.
- Click **■ Upload Document**. PDFs get an automatic thumbnail.

Finding Documents

- Use the Category filter, tag cloud, or sort options (Newest, Title, Most Downloaded).
- Switch between grid and list view using the view toggle.
- Click any document to open its detail, then Download to open, Edit to change metadata, or Delete.

TAB	CONTENTS
■ Amateur Radio	Radio manuals, frequency plans, ARRL publications, band plans, antenna references
■ ICS / Emergency Mgmt	SWIC plans, NIMS documents, ICS form packages, agency SOGs, mutual aid plans
■ All Documents	Every uploaded document regardless of category

WINLINK FORM IMPORT

<http://192.168.50.1/winlink-import.html>

32. Winlink Form Import & Incident Archiving

The Winlink Form Import page brings ICS form data from Winlink Express into the FieldComms incident record. When Winlink Express sends or receives an ICS form, it attaches an XML data file. This page parses that XML, maps the fields, and archives the form to the active incident.

Step-by-Step Import

- In Winlink Express, open the ICS form message (sent or received).
- Right-click the **RMS_Express_Form_*.xml** attachment and save it.
- On the Winlink Form Import page, drag the saved XML file onto the drop zone, or click Browse.
- Click **Parse Form Data**. The page detects the form type and extracts all fields.
- Review the extracted fields in the editable grid. Fix any fields that need correction.
- Choose the Incident and Direction (Received/Sent), then click **Archive to incident**.

FORM	RE-PRINT ON PI?	NOTES
ICS-213 General Message	Yes	Re-renders as the Pi ICS-213 for printing
ICS-214 Activity Log	Yes	Activity and personnel lines split into rows
ICS-309 Comms Log	Yes	Log rows parsed into the comms log
Other Winlink forms	Archive only	Data captured and archived — no ICS re-print

33

JS8CALL — HF DIGITAL MESSAGING

<http://192.168.50.1/>

33. JS8Call — HF Digital Keyboard Messaging

JS8Call is a keyboard-to-keyboard HF digital mode based on FT8 that provides store-and-forward messaging, heartbeats, and directed messaging at very weak signal levels. JS8Call runs on the Windows laptop connected to the IC-7300. The FieldComms dashboard provides a quick-launch card that connects to the JS8Call web API on the Windows machine.

Setup

- JS8Call must be running on the Windows laptop (not the Pi).
- Enable the JS8Call API: **File** → **Settings** → **API** — enable, set port to 2442.
- Ensure the Windows laptop is on EMCOMM-NET and note its IP address (e.g. 192.168.50.105).
- On the FieldComms dashboard, click the **JS8Call** card. Enter the Windows laptop IP when prompted.
- The card opens the JS8Call web interface directly in your browser.

SETTING	VALUE
Radio mode	USB-D (USB Digital)
USB cable	Single shielded USB-A to USB-B from IC-7300 to laptop
IC-7300 Modulation	USB-D, data off, AF level 50%, USB send enabled
JS8Call API port	2442 (default)
Recommended frequency	7.078 MHz USB-D (40m JS8Call calling frequency)

34

ICS PLANNING P

<http://192.168.50.1/ics/planningp.html>

34. ICS Planning P — Operational Planning Cycle

The Planning P page is an interactive guide to the ICS operational planning cycle. It presents all 15 phases from incident through each operational period, with the standard agenda, required ICS forms, and who should attend each phase. Access it from the Planning P tab on any ICS platform page.

How to Use

- Click the ■ **Planning P** tab from any ICS platform page.
- The reference diagram is shown on the left for orientation.
- Click any of the 15 phase buttons in the center panel.
- The right panel shows the standard agenda, required forms (as clickable links), and attendees.
- Click **Generate briefing sheet** to print a one-page phase cover for the IAP.

COLOR	GROUP	PHASES
Gray	Initial Response	1–5: Incident through Incident Brief
Yellow	Establish Objectives	6–7: IC/UC Objectives + C&G; Staff Meeting
Red	Develop the Plan	8–11: Tactics through Planning Meeting
Green	Prepare & Disseminate	12–13: IAP Prep + Operations Briefing
Teal	Execute, Evaluate & Revise	14–15: Execute Plan + New Ops Period

OPERATOR ACCESS CARDS

<http://192.168.50.1/printcenter.html>

35. Operator Access Cards

Operator Access Cards are printable Avery 5371 business-card-size reference cards (2" × 3.5", 10 per sheet). Each card shows the network name, server IP address, and quick-reference steps for connecting and getting started. Print, cut, laminate, and give one to every operator at a deployment.

Printing Operator Cards

- From the Print Center, click **Operator Access Cards**.
- The page generates a ready-to-print PDF formatted for Avery 5371 sheets.
- Print on cardstock or plain paper, cut along the card edges, and optionally laminate.

CARD FIELD	CONTENT
Network	EMCOMM-NET (Wi-Fi name to connect to)
Server address	http://192.168.50.1 (open in any browser)
Station callsign	K9ESV — MCESV/MCEMA
Steps	1. Connect to EMMCOMM-NET 2. Open browser 3. Go to 192.168.50.1
Identity	Enter your callsign/name on first use — remembered per browser

30. Quick Reference & Tips

Page Quick-Reference

TOOL	URL	WHEN TO USE
Dashboard	/	Opening screen — system status, weather alerts, APRS, tool access
Net Control	/netcontrol.html	Running any ARES/RACES amateur radio net
Starcom Logger	/starcom.html	Public safety Starcom nets with Radio IDs
ICS Platform	/ics/	Any formally-declared ICS incident
Tactical Map	/tactical.html	APRS situational awareness, unit tracking
Pre-Flight	/preflight.html	Before every activation — GO/CAUTION/NO-GO
Roster	/roster.html	Member management, activation tracking
NTS Radiogram	/nts.html	Generating and logging formal ARRL traffic
Kiwix Library	:8081/	Medical reference, equipment repair, field skills
HF Propagation	/propagation.html	Checking band conditions before HF operation
Repeaters	/repeaters.html	Finding local repeaters with ARES/SKYWARN support
Facilities	/facilities.html	Locating EOC, shelter, hospital, staging area
Print Center	/printcenter.html	Printing any form, ICS-309, or incident cover sheet
Cheat Sheets	/cheatsheets.html	Quick phonetic, Q-code, band plan, CTCSS reference
Dead Man's Switch	/deadmans.html	Monitoring net for inactivity during an activation

Field Operator Tips

- Keep EMCOMM-NET password posted at the EOC for new arriving operators.
- Bookmark <http://192.168.50.1/netcontrol.html> on your phone before the activation.
- The observer page works on any smartphone — share it with the Served Agency duty officer.
- Run the Pre-Flight checklist before every activation, not just major ones.
- The ICS-309 export from Net Control is the official communications record — export at the end of each operational period.
- Kiwix WikiMed is your best offline medical reference — faster than paper manuals.
- The Tactical Map auto-refreshes every 30 seconds. Leave it open on a dedicated tablet at the EOC.
- Use the Facilities page to keep EOC, shelter, and hospital frequencies current — update it before every deployment season.
- ARES/RACES members should have their certifications entered in the Roster before any activation.
- Export the Roster CSV monthly and keep a backup copy on the USB drive labeled FIELDCOMMS.

NOTE For questions or issues during an activation, the Health Monitor at <http://192.168.50.1:5051/health> shows complete system status. The install log is at `/var/log/fieldcomms-install.log`. Contact your net manager or Emergency Coordinator for operational questions.

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	8 GB LPDDR4X	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	TBD (current model)	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	✓ YES — Pi 5 chip, 8 GB RAM, current generation	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit (\$120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit (\$120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

501(c)(3) Grant Funding Note: MCEMV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	Pi 500 + Pi Monitor	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed
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■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.